

Not All Bad Demonstrations Are Equally Bad: Quantifying How Demonstration Failure Modes Degrade Closed-Loop Policy Performance

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<https://github.com/imjbassi/demo-quality-robustness>

September 2026

Abstract

Demonstration quality is often treated as a preprocessing concern in imitation learning. We instead hold the task, dataset size, and training protocol fixed while varying the type and prevalence of defective demonstrations. Five failure modes—occlusion, accidental success, corrective flailing, truncation, and inconsistent strategy—are introduced into a controlled 2D block-pushing task. We train 520 policies across ten paired training/evaluation seeds using two policy families, then run 120 additional transition-matched controls. For the MLP, full-contamination success is 0.965 for corrective flailing and 0.009 for accidental success, versus 0.949 for clean data. The paired changes are $+0.016 \pm 0.022$ and -0.940 ± 0.020 , respectively (95% t intervals). Accidental success remains catastrophic for Extra Trees (0.002), but the ordering is otherwise policy-dependent: full occlusion yields 0.866 success for the MLP and 0.990 for trees, whereas truncation yields 0.962 and 0.614. Pooled MLP open-loop error correlates strongly with closed-loop success ($r = -0.92$), yet within-mode correlations range from -0.13 to -0.95 ; 49 policies in the descriptive MSE band $[0.20, 0.30]$ span success from 0.130 to 0.995. Transition-matched controls confirm that the apparent inconsistent-strategy penalty is largely explained by alternate-strategy clonability rather than mixing itself. These results support defect- and policy-specific quality control and closed-loop validation, while remaining limited to simulation.

1 Introduction

Behaviour cloning is a standard baseline for robot manipulation [1], and its central known weakness is compounding error: small action errors move the policy off the demonstration distribution, where its predictions degrade further [2]. The standard responses to this are architectural or algorithmic: better policy classes [5], or interactive data collection.

Practitioners running data-collection pipelines know something the literature under-quantifies: the demonstrations themselves fail in structured, recognisable ways. A camera is occluded mid-episode. An operator fumbles and the object lands

on the goal anyway. Two operators solve the same task with incompatible strategies. An episode is cut short. These are not Gaussian noise on a clean signal; they are distinct defects with distinct structure, and quality-control effort spent on one is effort not spent on another.

Prior work has established that demonstration quality matters in aggregate [3, 4]. What is less established is a *ranking*: given finite QC budget, which defect should a pipeline prioritise? And, because closed-loop evaluation is expensive and open-loop action-prediction error is cheap, does the cheap metric even detect the damage?

This paper answers both questions in a deliberately small, fully controlled setting. Our contributions are:

1. A controlled design treating demonstration quality as the independent variable across five failure modes, ten paired seeds, and two policy families.
2. Evidence that the most damaging defect is robust across policies while the relative cost of occlusion and truncation is policy-dependent.
3. Evidence that a strong pooled association between open- and closed-loop metrics does not imply uniform diagnostic value within each failure mode.
4. Episode- and transition-matched controls showing that an apparent strategy-inconsistency effect is instead dominated by unequal clonability.

2 Method

2.1 Environment

Push2D is a 2D block-pushing task implemented in vectorised NumPy. A point gripper must move a block into a goal region. Pushing was chosen over reaching deliberately: it imposes a *staging* requirement, since the gripper must position itself behind the block relative to the goal before pushing is productive. This makes compounding error consequential. A policy that drifts during approach ends up on the wrong side and pushes the block *away* from the goal, which is unrecoverable within

the episode budget. This is precisely the class of failure that open-loop metrics are structurally poor at surfacing, since each individual action error is small.

Observations are 10-dimensional (gripper, block and goal positions plus two relative vectors); actions are 2D displacements clipped to $[-1, 1]$. Episodes run for at most 110 steps and terminate on success, defined as the block centre falling within 0.05 of the goal.

2.2 Scripted experts

Using scripted rather than human demonstrations is what makes the experiment controlled: corruption can be applied with exact, programmatic ground truth rather than human judgement about what counts as a bad demo.

The **primary expert** is a hand-coded orbit-then-push controller that circles the block at a safe radius until aligned, then pushes toward the goal. Measured success rate: **100.0%**.

The **alternate expert** always circles in the same rotational direction from a wider staging radius, producing a visibly different route through state space. It was explicitly tuned until it *also* reaches **100.0%**, so that mixing the two tests strategy *inconsistency* rather than strategy *quality*. This tuning turns out to matter substantially (Section 4).

The **naive controller** charges blindly at the block and never reasons about the goal; the block travels along whatever ray the gripper started on. It succeeds on approximately 6% of episodes, purely by geometric luck.

2.3 Failure modes

Each mode produces demonstrations that a real pipeline could plausibly log:

Occlusion. A contiguous window covering 35% of timesteps has its observations frozen at the last visible frame. Actions remain expert-quality; the demonstrator can see, the camera cannot. The resulting transitions have corrupted inputs and pristine labels.

Accidental success. Rollouts from the naive controller, keeping only those that reached the goal. Correct outcome, no correct behaviour anywhere in the episode.

Corrective flailing. Heavy uniform jitter on the first 12 timesteps, after which the demonstrator settles into correct motion and still succeeds.

Truncation. A clean episode cut at 70% of its length and logged anyway, what a QC filter missing a premature stop looks like.

Inconsistent strategy. Complete episodes from the alternate expert. Every one is a genuine success; the only defect is disagreement with the primary strategy about what to do in a given state.

2.4 Two design decisions

Action-level corruptions are re-simulated, not pasted. Overwriting actions in a logged trajectory without re-running the dynamics yields observations that no longer follow from the actions, a physically impossible episode. That measures

label noise, not bad demonstrations. Every mode that touches actions replays the episode through the environment.

Corrupted episodes must pass the same success-based QC filter a real pipeline applies. Modes are rejection-sampled to successful episodes only. Truncation is the deliberate exception, since it *is* the filter failing.

2.5 Sweep, policies, evaluation and uncertainty

The sweep axis is **contamination rate** ρ , not per-episode severity. Every dataset contains the same number of episodes; ρ of them are defective and $(1 - \rho)$ are clean, with per-episode severity held fixed. This places all five modes on one comparable axis and answers the question a pipeline actually asks: what fraction of the dataset can be this kind of bad before the policy breaks?

We sweep $\rho \in \{0, 0.25, 0.5, 0.75, 0.9, 1.0\}$ across ten training seeds. The shared clean baseline is fitted once per seed, producing 260 fits per policy family: ten clean fits and $5 \times 5 \times 10$ contaminated fits. The primary policy is a 2×256 ReLU MLP trained by MSE behaviour cloning for 50 epochs. A robustness sweep uses an Extra Trees regressor with 40 trees and a minimum leaf size of two [6]. Both train on the same 400-episode datasets.

Because alternate-strategy episodes are longer, we add a control that fixes the dataset at 13,000 transitions and defines ρ as the exact fraction of transitions drawn from that strategy. Repeating all six mixture levels for both policy families and ten seeds adds 120 fits, for 640 fits in total.

Two metrics are recorded per policy. **Open-loop MSE** is action-prediction error against 300 held-out *clean* expert episodes. **Closed-loop success rate** runs the policy over 200 episodes. Evaluation seed $12345 + s$ is used for training seed s ; evaluation states vary across seeds but are paired across configurations within a seed. We report means and two-sided 95% Student-t intervals across the ten paired seeds; full-contamination changes are paired against the corresponding clean fit.

3 Results

The MLP clean baseline reaches 0.949 ± 0.015 closed-loop success (mean and 95% t-interval half-width); the Extra Trees baseline reaches 0.996 ± 0.003 .

3.1 Failure modes are not interchangeable

Table 1 gives the headline comparison. At full contamination, corrective flailing changes MLP mean success by $+0.016 \pm 0.022$ relative to the paired clean baseline, whereas accidental success changes it by -0.940 ± 0.020 . Both are “bad demonstrations” under a practical taxonomy, yet their observed costs differ sharply.

Corrective flailing causes little degradation in this setting. Jitter-then-recover trajectories may provide DAGger-like coverage of off-distribution states [2]; at $\rho = 1.0$, MLP mean success is 0.965 ± 0.015 . The paired interval includes zero

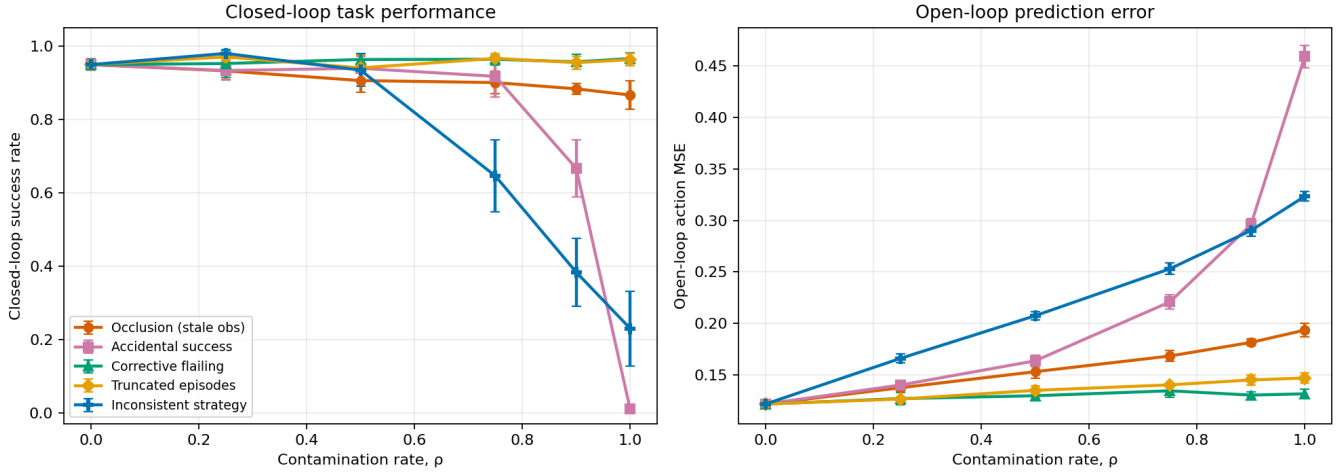


Figure 1: Dose-response curves across contamination rate. **Left:** closed-loop success rate, the quantity of interest. **Right:** open-loop MSE, the offline proxy. Accidental-success policies retain high mean success at $\rho = 0.75$ before collapsing at $\rho = 1.0$, whereas corrective flailing and truncation remain comparatively stable. Points are MLP means and error bars are 95% t intervals over ten paired seeds.

Failure mode	$\rho=0.5$	$\rho=1.0$	Pearson r
Corrective flailing	0.963	0.965	-0.26
Truncated episodes	0.940	0.962	-0.13
Occlusion (stale obs)	0.905	0.866	-0.64
Inconsistent strategy	0.934	0.229	-0.89
Accidental success	0.939	0.009	-0.95
Clean baseline	0.949		n/a

Table 1: MLP closed-loop success by failure mode and contamination rate, mean over ten seeds. Pearson r is the descriptive within-mode correlation between open-loop MSE and closed-loop success across 60 fits per mode.

change, so these data do not establish a benefit, but they make flailing a lower QC priority here.

Truncation does not reduce mean success in the observed setting: MLP mean success is 0.962 ± 0.017 at $\rho = 1.0$. A plausible explanation is that the push action can be extrapolated from the retained approach phase. This interpretation should not be transferred to tasks with a distinct terminal phase, such as grasping, without further experiments.

3.2 Accidental success is a cliff, not a slope

The most operationally significant observed result is the shape of the accidental success curve: 0.939 at $\rho = 0.5$, 0.917 at $\rho = 0.75$, 0.666 at $\rho = 0.9$, and **0.009** at $\rho = 1.0$.

In this experiment, a minority of clean demonstrations masks much of the problem. A pipeline monitoring only success-labelled demonstration throughput would not reveal the approaching policy failure. The diagnostic wrong-side push rate tells the same story, rising from 0.011 at $\rho = 0.5$ to 0.415 at $\rho = 1.0$: the policy has learned to shove the block in essentially arbitrary directions.

3.3 Open-loop evaluation has failure-mode-dependent value

Pooled across all 260 distinct MLP fits, open-loop MSE correlates with closed-loop success at $r = -0.92$. This strong aggregate association could appear to justify its use as a screening metric.

Broken out by failure mode, however, Pearson correlations range from -0.13 (truncation) to -0.95 (accidental success). The metric is strongly associated with success for some defects but weakly associated for truncation and corrective flailing, precisely where closed-loop performance varies little.

A complementary descriptive comparison considers the 49 runs whose open-loop MSE falls in the descriptive band [0.20, 0.30]. Their closed-loop success ranges from **0.130** to **0.995**. This band is not a statistical equivalence test; it illustrates that numerically similar offline scores need not determine similar task outcomes. Figure 2 shows the full distribution, including the high-MSE collapse points that contribute strongly to the pooled correlation.

4 Robustness controls

4.1 Policy family changes the ranking

Table 2 repeats the full-contamination comparison with Extra Trees. Accidental success remains catastrophic and corrective flailing remains benign. The middle of the ranking changes substantially: trees are robust to stale observations but sensitive to truncation, whereas the MLP shows the opposite pattern. Demonstration defects therefore interact with policy inductive bias; only the extreme cases are stable enough here to support a cross-policy priority.

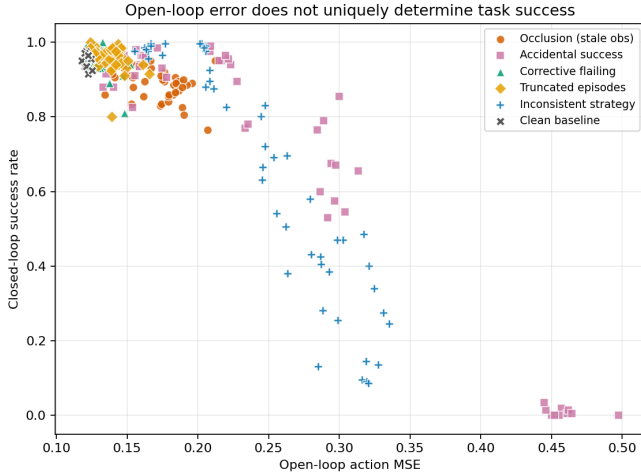


Figure 2: Open-loop MSE against closed-loop success for all 260 distinct MLP fits. Within the descriptive MSE band $[0.20, 0.30]$, closed-loop success spans 0.865.

Failure mode	MLP	Extra Trees
Corrective flailing	0.965 ± 0.015	0.993 ± 0.007
Truncated episodes	0.962 ± 0.017	0.614 ± 0.033
Occlusion (stale obs)	0.866 ± 0.038	0.990 ± 0.006
Inconsistent strategy	0.229 ± 0.102	0.142 ± 0.050
Accidental success	0.009 ± 0.008	0.002 ± 0.003

Table 2: Closed-loop success at $\rho = 1.0$ (mean $\pm$ 95% t-interval half-width over ten paired seeds).

4.2 Strategy mixing under transition matching

We initially predicted that mixing two valid strategies would cause multimodal-averaging collapse. The endpoint control contradicts that reading: at $\rho = 1.0$ the dataset is fully self-consistent, so its low success is a *clonability floor*, not evidence of mixing harm. Under episode matching, the MLP’s observed success exceeds the line between its $\rho = 0$ and $\rho = 1$ endpoints at every intermediate mixture.

The stricter transition-matched control fixes both total transitions and the alternate-strategy transition fraction. Its intermediate MLP means are 0.962, 0.966, 0.741, and 0.386; Extra Trees means are 0.984, 0.888, 0.797, and 0.563. Every point remains above the corresponding endpoint interpolation. Across both matching schemes, these curves support unequal clonability, rather than strategy conflict, as the dominant explanation; the complete control plot is included in the repository.

5 Limitations

Controlled simulation only. Push2D omits 3D geometry, contact uncertainty, visual perception, and hardware latency. The study tests a method for isolating defect effects, not transferable numerical thresholds.

Two deterministic regressors. The MLP and Extra Trees differ in inductive bias, but both return point actions. Multi-

modal generative policies, including diffusion policies [5], may respond differently to inconsistent strategies.

Finite seed and evaluation coverage. Ten paired seeds materially improve precision, and evaluation states vary by seed, but each fit is still tested on only 200 initial conditions from the same simulator distribution.

Exploratory analysis. The failure modes and MSE band were not preregistered, and intervals are not adjusted for multiple comparisons. They quantify run-to-run uncertainty rather than establish universal effects.

Scripted demonstrations. Programmatic defects provide exact control but do not capture the full diversity or temporal dependence of human data- collection errors.

6 Conclusion

Treating demonstration quality as the independent variable reveals both stable and policy-dependent effects. Accidental-success data is catastrophic near full contamination and corrective flailing is consistently benign, but occlusion and truncation exchange rank across the MLP and Extra Trees. The practical hypothesis is therefore interaction-aware: QC effort should depend on both defect type and the intended learner.

The evaluation result is equally important: open-loop action-prediction error has a strong pooled association with closed-loop success, yet its diagnostic value varies substantially by failure mode. The transition-matched analysis also turns an apparent inconsistency penalty into a clonability result. Offline screening and endpoint controls are useful, but neither substitutes for closed-loop validation.

Reproducibility statement

Source code, exact training and evaluation seeds, raw per-run results, and the scripts used to generate all three figures are available at <https://github.com/imjbassi/demo-quality-robustness>. The repository also records the software requirements and commands needed to reproduce all 640 policy fits and rebuild this manuscript.

References

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